

University of Cape Town
Department of Mathematics and Applied Mathematics

2IA Class Test 2
2nd October, 2023
Time: 2 hours
Total marks: 105
Full marks: 100

- Read the instructions carefully.
- You have 2hrs to complete the test.
- Make sure that no pages or questions are missing.
- Show all your work or reasoning. Answers without work or reasoning shown will receive no mark.

Surname: _____

First name: _____

Student number: _____

Signature: _____

Do not write below this line:

Question:	1	2	3	4	Total
Marks:	30	40	14	21	105
Score:					

1. (a) 5 marks State the Parity Theorem.
- (b) 3 marks Give an example of an even permutation.
- (c) Fix $n \geq 3$ (fixed). For $1 \leq i \neq j \leq n$, we write σ_{ij} to denote the transposition $(i \ j)$ (that is the permutation in S_n that swaps i and j and fixes any $k \in X_n$ different from i and j).
 - i. 7 marks Suppose that $1 \leq i < j \leq n$. Prove that

$$\sigma_{ij} = \sigma_{1i} \sigma_{1j} \sigma_{1i},$$
 that is, show that $(i \ j) = (1 \ i)(1 \ j)(1 \ i)$.
 - ii. 4 marks Show that any permutation in S_n can be written as a product of transpositions of the form $\sigma_{1i} = (1 \ i)$, for $2 \leq i \leq n$.
 - iii. 7 marks For $2 \leq i \neq j \leq n$, prove that $(1 \ i \ j) = \sigma_{1j} \sigma_{1i} = (1 \ j)(1 \ i)$.
 - iv. 4 marks Prove that any even permutation can be written as a product of cycles of the form $(1 \ i \ j)$, where $2 \leq i \neq j \leq n$.

Solution:

(a) PARITY THEOREM. If a permutation $\sigma \in S_n$ admits two different factorisations onto transpositions:

$$\sigma = \gamma_s \dots \gamma_2 \gamma_1 = \mu_r \dots \mu_2 \mu_1, \text{ (here } \gamma_i \text{ and } \mu_j \text{ are transpositions)}$$

Then r and s have the same parity; that is, either r and s are even, or r and s are odd.

(b) The identity is an even transposition, as well, as the 3-cycle $(1 \ 2 \ 3)$, since $(1 \ 2 \ 3) = (1 \ 2)(2 \ 3)$.

(c) i. We need to prove that $\sigma_{ij}(k) = (\sigma_{1i} \circ \sigma_{1j} \circ \sigma_{1i})(k)$ for all $k \in X_n = \{1, 2, \dots, n\}$. We distinguish a few cases:

- If $k \notin \{i, j\}$, then

$$\begin{aligned} \sigma_{ij}(k) &= k, \\ (\sigma_{1i} \circ \sigma_{1j} \circ \sigma_{1i})(k) &= (\sigma_{1i} \circ \sigma_{1j})(k) = \sigma_{1i}(k) = k. \end{aligned}$$

- For $k = i$, we have that

$$\begin{aligned} \sigma_{ij}(i) &= j, \\ (\sigma_{1i} \circ \sigma_{1j} \circ \sigma_{1i})(i) &= (\sigma_{1i} \circ \sigma_{1j})(1) = \sigma_{1i}(j) = j. \end{aligned}$$

- For $k = j$, we get

$$\begin{aligned} \sigma_{ij}(j) &= i, \\ (\sigma_{1i} \circ \sigma_{1j} \circ \sigma_{1i})(j) &= (\sigma_{1i} \circ \sigma_{1j})(j) = \sigma_{1i}(1) = i. \end{aligned}$$

ii. Applying the cycle decomposition theorem, we can factorise any permutation σ different from the identity as a product of disjoint cycles. On the other hand, any cycle $(k_1 \ k_2 \ \dots \ k_r)$ can be factorised onto a product of transpositions:

$$(k_1 \ k_2 \ \dots \ k_{r-1} \ k_r) = (k_1 \ k_2)(k_2 \ k_3) \dots (k_{r-1} \ k_r) = \sigma_{k_1 k_2} \sigma_{k_2 k_3} \dots \sigma_{k_{r-1} k_r}.$$

Putting these together, the result follows now from (i).

iii. Let $\alpha_{1ij} = (1 \ i \ j)$. We proceed like in (i) to prove that $\alpha_{1ij} = \sigma_{1j}\sigma_{1i}$. For $k \in X_n = \{1, 2, \dots, n\}$. We distinguish a few cases:

- If $k \notin \{i, j\}$, then

$$\begin{aligned} \alpha_{1ij}(k) &= k, \\ (\sigma_{1j} \circ \sigma_{1i})(k) &= \sigma_{1j}(\sigma_{1i}(k)) = \sigma_{1j}(k) = k. \end{aligned}$$

- For $k = 1$, we have that

$$\begin{aligned} \alpha_{1ij}(1) &= i, \\ (\sigma_{1j} \circ \sigma_{1i})(1) &= \sigma_{1j}(\sigma_{1i}(1)) = \sigma_{1j}(i) = i. \end{aligned}$$

- For $k = i$, we have that

$$\begin{aligned} \alpha_{1ij}(i) &= j, \\ (\sigma_{1j} \circ \sigma_{1i})(i) &= \sigma_{1j}(\sigma_{1i}(i)) = \sigma_{1j}(1) = j. \end{aligned}$$

- For $k = j$, we get

$$\begin{aligned} \alpha_{1ij}(j) &= 1, \\ (\sigma_{1j} \circ \sigma_{1i})(j) &= \sigma_{1j}(\sigma_{1i}(j)) = \sigma_{1j}(j) = 1. \end{aligned}$$

iv. Let σ be an even permutation. From (ii) we can conclude that σ can be written as a product of transpositions of the form σ_{1k} . Using the parity Theorem we know that we will have an even number of σ_{1k} in such factorisation. The result now follows from (iii).

2. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**

- (a) 5 marks The set of all permutations S_3 is an abelian group.
- (b) 5 marks The permutation $(1\ 2)(2\ 3)\dots(n-2\ n-1)(n-1\ n) \in S_n$ is a cycle.
- (c) 5 marks If $\alpha, \beta \in S_n$, then the permutation $\sigma = \alpha^{-1}\beta^{-1}\alpha\beta$ is always even.
- (d) 5 marks If G is a group, then the empty set is always a subgroup of G .
- (e) 5 marks Every abelian group is cyclic.
- (f) 5 marks Every group of order 4 is cyclic.
- (g) 5 marks Let a be an element of a group G . If $a^n = a^m = 1$ for some coprime integers n and m , then $a = 1$.
- (h) 5 marks If H and K are two subgroups of a group G , then their union $H \cup K$ is also subgroup of G .

Solution: (a) **False:** The transpositions $\alpha = (1\ 2)$ and $\beta = (1\ 3)$ are in S_n and they do not commute. In fact:

$$\begin{aligned}\alpha\beta &= (1\ 2)(1\ 3) = (1\ 3\ 2), \\ \beta\alpha &= (1\ 3)(1\ 2) = (1\ 2\ 3)\end{aligned}$$

(b) **True:** We proved in the workshops that for any r -cycle $(k_1\ k_2\ \dots\ k_r)$ we have the following factorisation onto transpositions:

$$(k_1\ k_2\ \dots\ k_r) = (k_1\ k_2)(k_2\ k_3)\dots(k_{r-1}\ k_r).$$

Using this we obtain that

$$(1\ 2)(2\ 3)\dots(n-2\ n-1)(n-1\ n) = (1\ 2\ 3\ \dots\ n-1\ n),$$

which is a cycle.

(c) **True:** Let us begin by noticing that the inverse σ^{-1} of a permutation σ has the same parity that σ . We distinguish two cases:

- Case 1: α and β are both odd:

Then α^{-1} and β^{-1} are odd, and so $\alpha^{-1}\beta^{-1}\alpha\beta$ is even (we will have an even number of transpositions in its factorisation; odd + odd + odd + odd = even).

- Case 2: Either α is even or β is even:

Then either α^{-1} or β^{-1} is even. In any case we get that $\alpha^{-1}\beta^{-1}\alpha\beta$ is even (either we would have even + even + even + even, or odd + even + odd + even).

(d) **False:** The empty group can **never** be a subgroup of a subgroup since it does not contain the identity element of the group.

(e) **False:** For instance, $\mathbb{Z}_2 \times \mathbb{Z}_2$ is abelian but it is not cyclic, since every element of $\mathbb{Z}_2 \times \mathbb{Z}_2$ has order 2 and $|\mathbb{Z}_2 \times \mathbb{Z}_2| = 4$.

(f) **False:** We can use the same example that in (e). $\mathbb{Z}_2 \times \mathbb{Z}_2$ has order 4 but it is not cyclic, since every element of $\mathbb{Z}_2 \times \mathbb{Z}_2$ has order 2.

(g) **True:** Apply Bézout's Lemma to write $1 = xn + ym$ for some integers x and y . From here, using the exponent laws together with the fact that $a^n = a^m = 1$ we get that

$$a = a^1 = a^{xn+ym} = (a^n)^x \cdot (a^m)^y = 1,$$

as required.

(g) **False:** Take $G = \mathbb{Z}$ and $H = 2\mathbb{Z}$ and $K = 3\mathbb{Z}$. Then notice that $H \cup K$ is not a subgroup of G since $2 \in H$, $3 \in K$ but $2 + 3 = 5 \notin H \cup K$.

3. Consider the additive cyclic group $\mathbb{Z}_{30}$.

- (a) 5 marks Find all the subgroups of $\mathbb{Z}_{30}$.
- (b) 6 marks For each of the subgroups of $\mathbb{Z}_{30}$, write down all its elements.
- (c) 3 marks Draw the lattice diagram of $\mathbb{Z}_{30}$.

Solution: (a) and (b). The positive divisors of 30 are 1, 2, 3, 5, 6, 10, 15, 30. Thus, the Fundamental Theorem of Finite Cyclic Groups tells us that $\mathbb{Z}_{30}$ has exactly 8 subgroups, exactly one per positive divisor of 30. It is clear that the subgroups of orders 1 and 30 are $\{\bar{0}\}$ and $\mathbb{Z}_{30}$, respectively. Moreover, for $k \in \{2, 3, 5, 6, 10, 15\}$, applying the Fundamental Theorem of Finite Cyclic Groups we obtain that $\langle \frac{30}{k}\bar{1} \rangle$ is the unique subgroup of $\mathbb{Z}_{30}$ of order k .

Using this, we obtain that

$$\text{order 2: } \left\langle \frac{30}{2}\bar{1} \right\rangle = \langle \bar{15} \rangle = \{\bar{0}, \bar{15}\},$$

$$\text{order 3: } \left\langle \frac{30}{3}\bar{1} \right\rangle = \langle \bar{10} \rangle = \{\bar{0}, \bar{10}, \bar{20}\},$$

$$\text{order 5: } \left\langle \frac{30}{5}\bar{1} \right\rangle = \langle \bar{6} \rangle = \{\bar{0}, \bar{6}, \bar{12}, \bar{18}, \bar{24}\},$$

$$\text{order 6: } \left\langle \frac{30}{6}\bar{1} \right\rangle = \langle \bar{5} \rangle = \{\bar{0}, \bar{5}, \bar{10}, \bar{15}, \bar{20}, \bar{25}\},$$

$$\text{order 10: } \left\langle \frac{30}{10}\bar{1} \right\rangle = \langle \bar{3} \rangle = \{\bar{0}, \bar{3}, \bar{6}, \bar{9}, \bar{12}, \bar{15}, \bar{18}, \bar{21}, \bar{24}, \bar{27}\},$$

$$\text{order 15: } \left\langle \frac{30}{15}\bar{1} \right\rangle = \langle \bar{2} \rangle = \{\bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \bar{10}, \bar{12}, \bar{14}, \bar{16}, \bar{18}, \bar{20}, \bar{22}, \bar{24}, \bar{26}, \bar{28}\}.$$

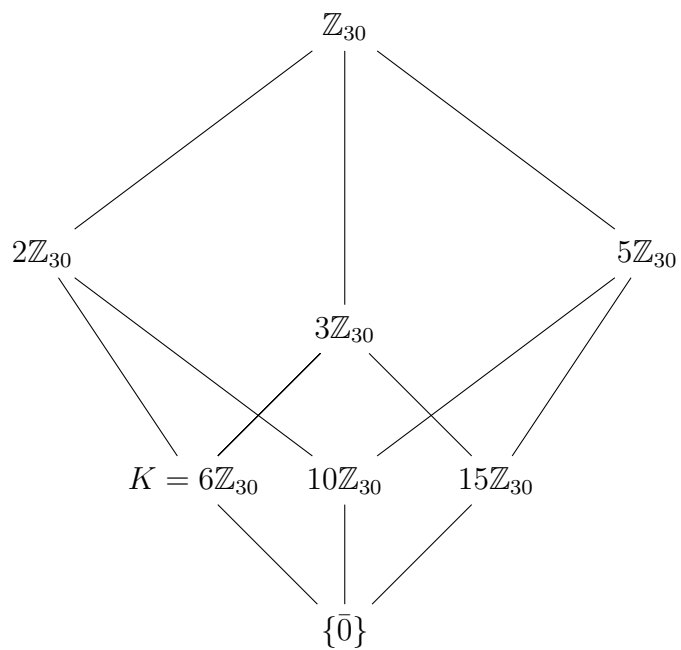
(c) Notice that

$$\langle \bar{15} \rangle \subseteq \langle \bar{3} \rangle \quad \text{and} \quad \langle \bar{15} \rangle \subseteq \langle \bar{5} \rangle,$$

$$\langle \bar{10} \rangle \subseteq \langle \bar{5} \rangle \quad \text{and} \quad \langle \bar{10} \rangle \subseteq \langle \bar{2} \rangle,$$

$$\langle \bar{6} \rangle \subseteq \langle \bar{2} \rangle, \quad \text{and} \quad \langle \bar{6} \rangle \subseteq \langle \bar{3} \rangle.$$

Using these we get the following lattice diagram:



4. (a) 6 marks Complete the missing parts (marked with a $\star$) in the statement of the theorem that describes the cyclic subgroup generated by an element.

THEOREM. Let g be an element of a group G .

- (i) **If the cyclic subgroup $\langle g \rangle$ is finite**, then there exists a $(\star)$ n such that $(\star)$. In such a case:
- (a) $g^k = 1$ if and only if $(\star)$.
 - (b) $g^k = g^m$ if and only if $(\star)$.
 - (c) $\langle g \rangle = \{1, g, g^2, \dots, g^{n-1}\}$ and the elements $1, g, g^2, \dots, g^{n-1}$ are all distinct.
- (ii) **If the cyclic subgroup $\langle g \rangle$ is infinite**, then:
- (d) $g^k = 1$ if and only if $(\star)$.
 - (e) $g^k = g^m$ if and only if $(\star)$.
 - (f) $\langle g \rangle = \{\dots, g^{-2}, g^{-1}, 1, g, g^2, \dots\}$ and all these powers of g are distinct.

- (b) Consider the following incomplete proof of (i) (a) in the theorem above:

(i) Suppose that $\langle g \rangle$ is finite. Then the powers $g, g^2, g^3, \dots$ can not be all distinct, so we can find two distinct positive integers k and m such that $g^k = g^m$. Assume, for example, that $k < m$. Then $g^{m-k} = 1$, where $m - k > 0$, so by the *Well-Ordering Axiom* there exists a smallest positive integer n such that $g^n = 1$.

(a) Assume first that $n|k$. Then (A). Thus:

(B)

Conversely, if $g^k = 1$, applying the (C) we can write $k = qn + r$ with $0 \leq r < n$. Then

(D)

which implies (E) and so $n|k$.

With reference to this partial proof:

- i. 2 marks Complete the sentence at (A)
- ii. 5 marks Fill in the necessary steps at (B) to conclude that $g^k = 1$.
- iii. 2 marks Name the result you are using in (C).
- iv. 6 marks Fill in the necessary steps at (D) and (E) to conclude that $n|k$.

Solution:

(a) THEOREM. Let g be an element of a group G .

(i) **If the cyclic subgroup $\langle g \rangle$ is finite**, then there exists a **a smallest positive integer n** such that **$g^n = 1$** . In such a case:

(a) $g^k = 1$ if and only if **$n|k$** .

(b) $g^k = g^m$ if and only if **$k \equiv m \pmod{n}$** .

(c) $\langle g \rangle = \{1, g, g^2, \dots, g^{n-1}\}$ and the elements $1, g, g^2, \dots, g^{n-1}$ are all distinct.

(ii) **If the cyclic subgroup $\langle g \rangle$ is infinite**, then:

(d) $g^k = 1$ if and only if **$k = 0$** .

(e) $g^k = g^m$ if and only if **$k = m$** .

(f) $\langle g \rangle = \{\dots, g^{-2}, g^{-1}, 1, g, g^2, \dots\}$ and all these powers of g are distinct.

(b)

(A) $k = qn$ for some integer $q \in \mathbb{Z}$.

(B) $g^k = g^{qn} = (g^n)^q = 1$.

(C) Division Algorithm.

(D) $g^r = g^{(k - qn)} = g^k (g^n)^{-q} = 1$.

(E) $r = 0$, since n was the least positive integer with that property.